

Detailed Simulation Model: Supplementary Material for WSC 2026

Phu Nguyen^a, Dmitry Ivanov^b

^a*Department of Business and Economics, Berlin School of Economics and Law,
Berlin 10825, Germany
phu.nguyen@hwr-berlin.de*

^b*Department of Business and Economics, Berlin School of Economics and Law,
Berlin 10825, Germany
dmitry.ivanov@hwr-berlin.de*

Abstract

This document describes the simulation model and the underlying data used to illustrate the effectiveness of the adaptive simulation-surrogate framework.

1. Simulation model

The DES mimics a make-to-order supply chain and includes three major processes: demand management, material procurement, and production and fulfillment. Figure 1 illustrates the interactions between the focal manufacturer, its customers, and its suppliers.

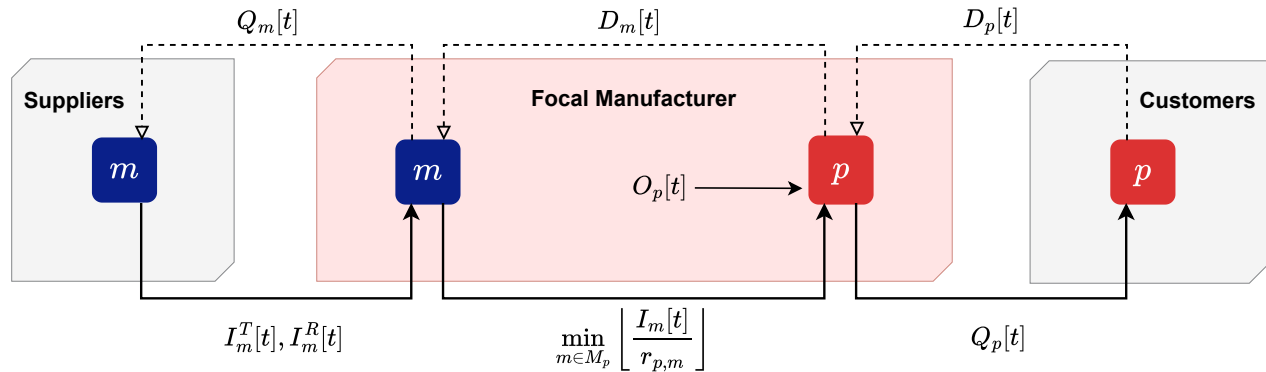


Figure 1: Illustration of supply chain DES model.

Product demand management. Each week (t), the focal company receives a requested delivery schedule from its customers for each product $p \in P$, where P denotes the finite set of all products. The schedule specifies the quantities of product p expected to be delivered in future periods. The primary motivation for customers to provide such schedules is the long lead time associated with material procurement. Operationally, at the beginning of week t , the focal company receives forward demand over a horizon of length τ^* , denoted $D_p[t + \tau^*]$, for each product p . During week t , the focal company produces to fulfil the current demand $D_p[t]$, while leveraging the visibility of $D_p[t + 1, t + \tau^*]$ to support material requirements planning and procurement decisions. For simplicity, a uniform visibility horizon of 52 weeks is assumed across all products.

Material procurement. To prepare for future demand, the focal manufacturer needs to release purchasing orders in advance. There are three sub-steps in the procurement process: material demand calculation, supplier selection and purchasing request release, and inventory controlling.

The focal company calculates the projected material requirements using the bill of materials (BoM) data. For each product $p \in P$ and material $m \in M$ (M is the set of all required materials), let $r_{p,m}$ denote the BoM consumption rate, the number of units of material m required to manufacture one unit of product p . Let $D_m[t]$ denote the corresponding material demand for material m in the same week. The total projected requirement for material m in week t is then computed as Eq. (1).

$$D_m[t] = \sum_{p \in P} D_p[t] \cdot r_{p,m} \quad (1)$$

Once the demand for material is projected, the company identifies a preferred supplier for each material $m \in M$. The sourcing database provides a list of potential suppliers $\mathcal{S}_m$ for each material, including the ordering cost per unit (c_o), the minimum order quantity (MOQ, denoted as Q_{MOQ}), and the supplier lead time (T_s). The focal manufacturer prioritize the lowest material cost when there are multiple suppliers. The choice of supplier is also important for inventory control policies. The company apply the min-max policies to control inventory. The min (s_m) and max (S_m) inventory levels are set in Eqs. (2) and (3) respectively. We assume κ is 8 weeks for all materials.

$$s_m[t] = \sum_{\tau=t}^{t+T_s} D_m[\tau] \quad (2)$$

$$S_m[t] = \sum_{\tau=t}^{t+T_s+\kappa} D_m[\tau] \quad (3)$$

An order is triggered when the total available inventory, comprising on-hand (I_m), in-transit (I_m^T), and scheduled receipts (I_m^R), falls below the reorder point s_m . The order quantity (Q_m) in week t is then computed as shown in Eq. (4).

$$Q_m[t] = \begin{cases} \max(S_m[t] - (I_m[t-1] + I_m^T[t] + I_m^R[t]), Q_{\text{MOQ}}), & \text{if } I_m[t-1] + I_m^T[t] + I_m^R[t] < s_m[t] \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

Once purchasing order are released, we update the inventory position accordingly for all the material using Eqs. (5) - (6).

$$I_m^T[t + \tau] = I_m^T[t + \tau] + Q_m[t], \quad \forall \tau \in [1, T_s - 1] \quad (5)$$

$$I_m^R[t + T_s] = I_m^R[t + T_s] + Q_m[t] \quad (6)$$

Production and demand fulfillment. At the beginning of each week t , the company takes a snapshot of the available inventory for each material $m \in M$ by calculating the on-hand inventory. This value is defined as the sum of the on-hand inventory from the previous week and the scheduled receipts arriving in the current week, as shown in Eq. (7).

$$I_m[t] = I_m[t-1] + I_m^R[t] \quad (7)$$

For each product $p \in P$, the maximum producible quantity in week t is constrained by the

availability of required materials. Let M_p denote the set of materials needed to produce product p . The actual production quantity $Q_p[t]$ is further bounded by the product-specific production capacity $O_p[t]$ and the product demand $D_p[t]$, as shown in Eq. (8).

$$Q_p[t] = \min \left(D_p[t], O_p[t], \min_{m \in M_p} \left\lfloor \frac{I_m[t]}{r_{p,m}} \right\rfloor \right) \quad (8)$$

Once the production quantity is determined, we update the corresponding material consumption from the inventory snapshot of week t for each material $m \in M_p$ as Eq. (9) describes.

$$I_m[t] = I_m[t] - Q_p[t] \cdot r_{p,m} \quad (9)$$

We employ fill rate as the primary indicator to measure supply chain performance and identify steady state Ivanov (2025). Fill rate is defined as the ratio of fulfilled demand to total demand, weighted by financial value, where u_p denotes the unit price of product p (see Eq. (10)). We select fill rate for steady-state detection because it directly measures system output performance normalized by demand. Unlike absolute delivery quantities (i.e., revenue), which fluctuate with demand variation even in steady state, fill rate captures the ability of the supply chain system to satisfy demand and therefore provides a clearer signal of steady-state convergence.

$$FR[t] = \sum_{p \in P} \frac{Q_p[t]}{D_p[t]} \cdot u_p \quad (10)$$

Figure 2 captures the baseline model of fill rate with 30 replications over 156 weeks (3 years). To ensure valid steady-state inference, we applied two warmup detection methods and selected the more conservative result. Conway's rule suggests the steady-state starts after week 84 (Conway, 1963), while the MSER-5 (Marginal Standard Error Rule with batch size of 5 data points) procedure indicates steady-state after week 15 (Hoad et al., 2010; White Jr, 1997).

We therefore define week 85 as the start of steady state for our supply chain system. The simulation analysis period spans from week 85 to week 137 (52 weeks), and all experiments are conducted using 30 independent random seeds.

Supply chain disruptions are modeled either as time delays or as capacity reductions (Namdar et al., 2022). In this study, we focus on deep disruptions at suppliers. To capture such disruptions, we explicitly represent supplier disruption as time delays in inbound logistics processes (Talluri et al., 2013; Namdar et al., 2022).

Grounded on the structured experiment design framework, we describe a disruption event using three parameters: disruption location, disruption start time, and disruption duration (Macdonald et al., 2018). A disruption event changes the stable state of the supply chain model as follows. Suppose a disruption occurs at supplier s^* at time $t = t^*$ with a duration of Δt week(s), the lead time for fulfilling material orders during the disruption period increases to $T_{s^*}^{disr} = T_{s^*} + \Delta t$. Therefore, all scheduled receipts originally planned within the disruption interval $t \in [t^*, t^* + \Delta t]$ are delayed by Δt . The scheduled receipts for the affected material m^* are thus updated as Eq. (11) describes.

$$\begin{aligned} I_{m^*}^{R,disr}[t + \Delta t] &= I_{m^*}^R[t], \\ I_{m^*}^{R,disr}[t] &= 0 \end{aligned} \quad (11)$$

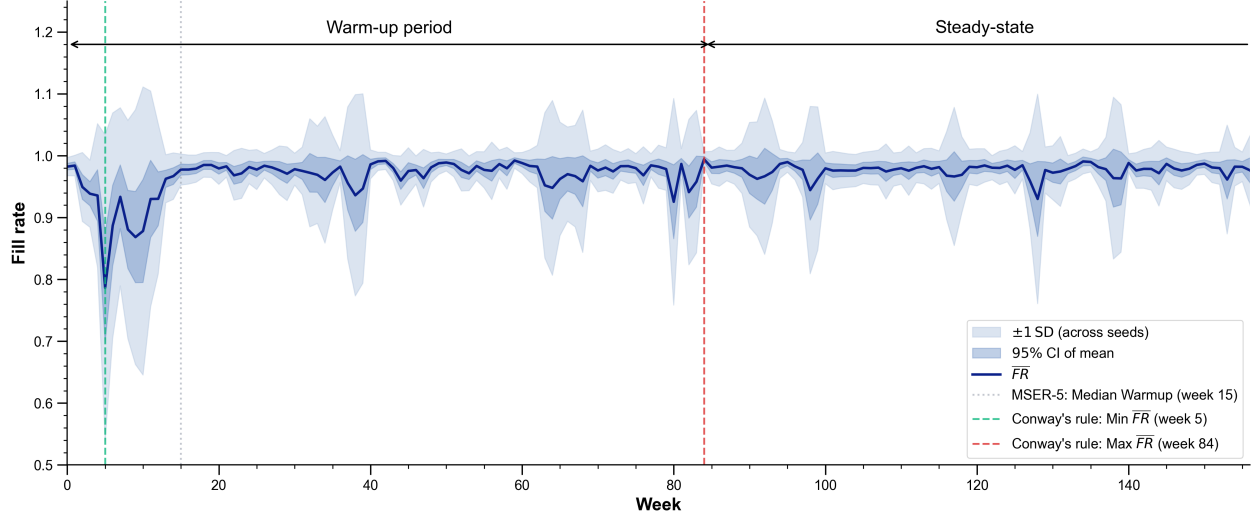


Figure 2: Average weekly fill rate across seeds and steady-state identification.

Furthermore, in-transit inventory existing at disruption time window, i.e., $I_m^T[t] > 0$, is also propagated forward across the disruption horizon (see Eq. (12)).

$$I_{m^*}^{T,disr}[t + \tau] = I_{m^*}^T[t], \quad \forall \tau \in [1, \Delta t - 1]. \quad (12)$$

In our study, the disruption duration is set between five and ten weeks using an uniform distribution. The disruption start time varies between weeks 85 and 87, also with a uniform probability distribution. We conduct a stress test by disrupting individual suppliers to identify the most vulnerable suppliers using the disruption profile described above. The stress test results are summarized in Figure 3.

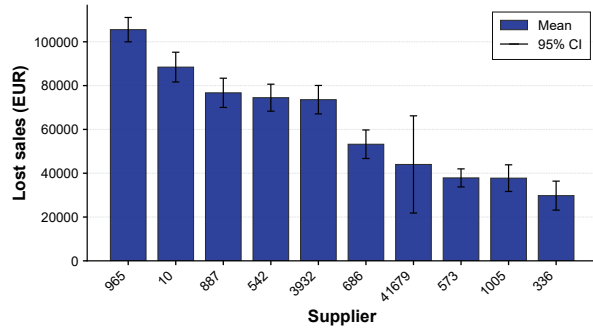


Figure 3: Mean revenue loss across top 10 most impact suppliers over 30 random seeds.

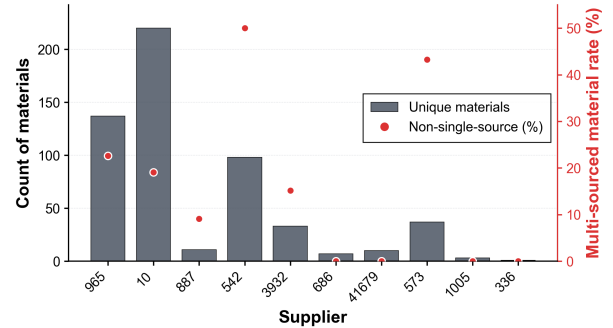


Figure 4: Material count and non-single-source material rate for top 10 most impact suppliers.

2. Details of products and supply chain model assumptions

Table 1 presents structural characteristics of the 15 products, including BOM complexity, supply base size, supplier lead times, and weekly demand distributions. Table 2 summarizes core modeling assumptions in this study.

Table 1: Product-Level Characteristics and Supply Chain Attributes

Product	BOM Complexity <i>Materials (n)</i>	Supply Base Size <i>Suppliers (n)</i>	Supplier Lead Time (weeks, across suppliers) <i>Min / Mode / Max</i>	Demand Distribution (units/week) <i>Min / Mode / Max</i>
Product 1	14	5	4.0 / 9.0 / 21.0	6.0 / 7.0 / 8.0
Product 2	95	17	3.0 / 11.0 / 36.0	10.0 / 17.0 / 30.0
Product 3	136	17	3.0 / 11.0 / 36.0	11.0 / 18.0 / 30.0
Product 4	5	4	3.0 / 13.0 / 31.0	8.0 / 28.0 / 72.0
Product 5	4	2	11.0 / 12.0 / 13.0	5.0 / 12.0 / 25.0
Product 6	156	12	2.0 / 15.0 / 31.5	7.0 / 9.0 / 10.0
Product 7	9	6	5.0 / 12.5 / 27.0	7.0 / 12.0 / 20.0
Product 8	4	4	1.0 / 11.0 / 21.0	8.0 / 14.0 / 24.0
Product 9	12	6	2.0 / 13.5 / 21.0	4.0 / 5.0 / 5.0
Product 10	26	7	7.0 / 13.0 / 33.0	4.0 / 7.0 / 10.0
Product 11	95	15	3.0 / 8.0 / 51.0	8.0 / 18.0 / 36.0
Product 12	7	5	4.0 / 9.0 / 14.0	15.0 / 30.0 / 60.0
Product 13	8	7	3.0 / 10.0 / 18.0	7.0 / 9.0 / 10.0
Product 14	7	6	7.0 / 15.0 / 48.0	26.0 / 46.0 / 80.0
Product 15	15	12	1.0 / 6.0 / 18.0	7.0 / 21.0 / 50.0

Note: All data are derived from a real-world supply chain, except for the min of product demand. Supplier lead times are observed at the *material-supplier* level, with each material associated with a supplier-specific lead time. Demand distribution parameters are derived from historical weekly demand. In the simulation, weekly demand for each product p is sampled independently as $D_p[t] \sim \text{Triangular}(a_p, b_p, c_p)$, where b_p is the historical median of weekly demand for product p , c_p is the historical maximum weekly demand, and $a_p = \max\{0, (1 - \nu)b_p\}$ with $\nu = 0.30$. The estimation of the min demand (a_p) accommodates the partial observability due to modeling a subnetwork of a larger supply chain.

Table 2: Summary of key supply chain model assumptions.

Parameter / Description	Value
<i>Planning and Inventory Control</i>	
Demand visibility horizon	52 weeks
Protective period (κ)	8 weeks
<i>Simulation Configuration</i>	
Total simulation horizon	156 weeks
Independent replications	30 seeds
Time discretization	Weekly
<i>Disruption Scenario</i>	
Disruption start time (t^*)	Uniform{86, 87, 88}
Disruption duration (Δt)	Uniform{5, 6, 7, 8, 9, 10}
<i>Resilience Strategy Parameters</i>	
Backup supplier lead time (T_s)	6 weeks
Annual inventory holding cost (h_m)	20% of material cost
Overtime production cost (C_p^o)	5% of product price
Material allocation horizon	4 weeks
Expedited shipment cost (C_m^{exp})	3% of material cost per order
Annual labor cost for material allocation planning (C^{alc})	6,240

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